

29. Ans: A

Let the distance be d .

At distance d , the intensity $I = P/4\pi d^2$

The power received by the telescope $P_{rec} = I \times \text{Area of aperture}$
 $= PA/4\pi d^2$

$$\text{Also } P_{rec} = \frac{N}{t} hf$$

$$\frac{PA}{4\pi d^2} = \frac{N}{t} hf$$

$$\text{Hence } \frac{10 \times 2.00 \times 10^{-3}}{4\pi d^2} = \frac{50}{0.10} \times 6.63 \times 10^{-34} \times 6.20 \times 10^{14}$$

$$d = 2.78 \times 10^6 \text{ m}$$